Module - II : Diffraction of X-rays

Introduction

X-rays are electro-magnetic wave with very short wavelength (0.1 – 100 Å). They have high penetrating power. X-rays are produced when fast electron incidents on matter.

Laue's Experiment

For diffraction of X-rays, the size of slit should be comparable to the wavelength of X-ray. Therefore, X-ray diffraction is not possible with grating surfaces. Von Laue suggested that X-ray diffraction can be observed by using crystals because crystals are regular arrangements of atoms with inter-atomic spacing of the order of 10^{-10} m.

Experimental Arrangement:

X-rays are produced by coolidge tube. They are passed through fine holes into lead screens S_1 & S_2. A fine beam of X-rays emerges out and incident on a zinc sulphide crystal. The diffracted beam is collected on a photo-graphic plate.

Observations:

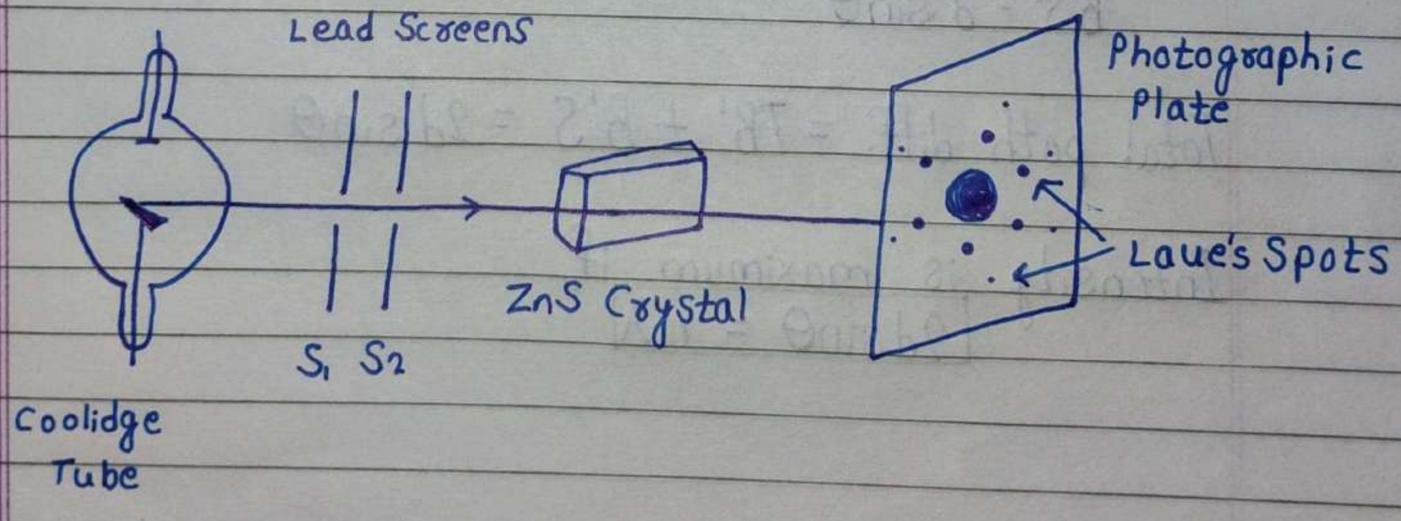
Plate was exposed to X-ray for 40 minutes. After developing, it was found that there was a big bright centre spot. This spot is surrounded by a number of symmetrically arranged spots called Laue's spot. Each spot corresponds to a particular angle of diffraction.

Imp. Outcomes:

1. X-rays are electro-magnetic wave with very short wavelength.

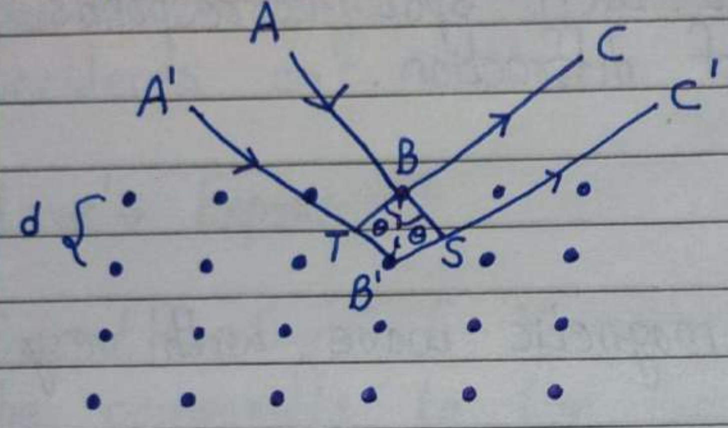
2. Diffraction of X-rays is possible by passing through crystal.

3. The atoms in a crystal are arranged in regular 3D array with inter-atomic distance of the order of 1 Å.



Bragg's Law

When parallel monocromatic X-ray beam incident on a crystal it is reflected. According to Bragg's Law, intensity of reflected beam is maximum if path difference is $n\lambda$.



In $BB'T$,

$$\sin\theta = \frac{TB'}{BB'} = \frac{TB'}{d}$$

$$\Rightarrow TB' = d\sin\theta$$

Similarly, in $BB'S$,

$$B'S = d\sin\theta$$

Total path diff. $= TB' + B'S = 2d\sin\theta$

Intensity is maximum if,

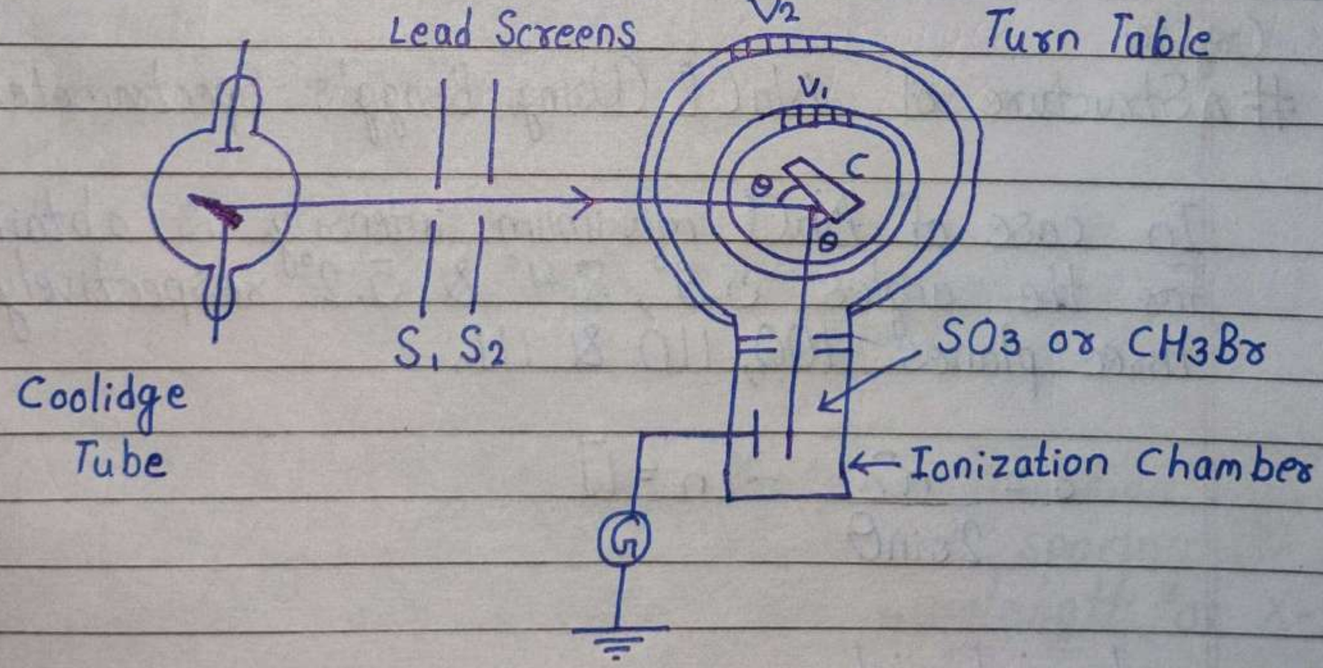
$$\boxed{2d\sin\theta = n\lambda}$$

Bragg's Spectrometer

Experimental Arrangement:

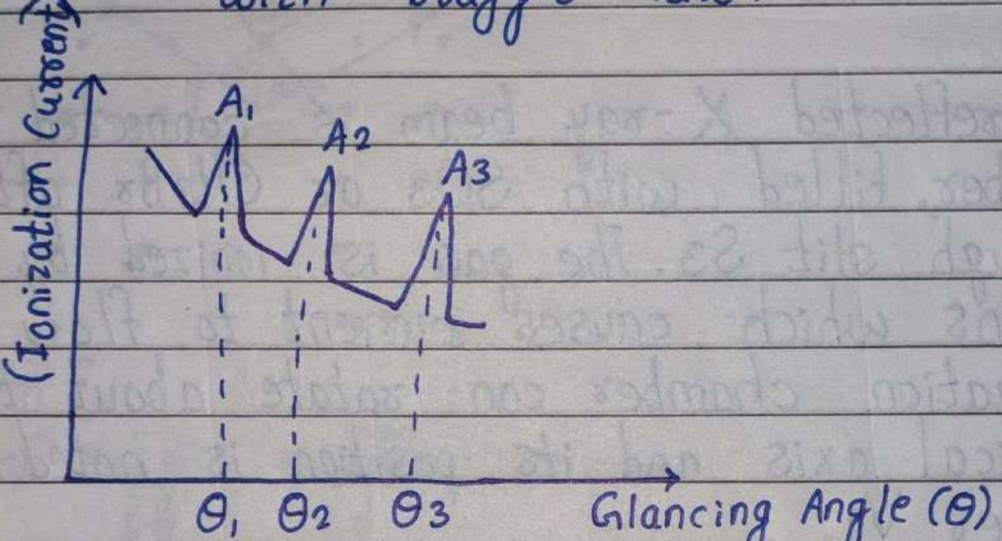
The crystal 'C' is mounted on a prism table which can rotate about a vertical axis. Its position can be noted by a vernier V_1. Beam of X-ray is incident on crystal after passing through lead screens S_1 & S_2.

The reflected X-ray beam is connected in ionization chamber, filled with SO_3 or CH_3Br after passing through slit S_3. The gas is ionized by X-ray photons which causes current to flow in galvanometer. Ionization chamber can rotate about the same vertical axis and its position is noted by another vernier V_2.



Observation:

The ionization chamber is adjusted to get maximum intensity. A graph is plotted between glancing angle and ionization current. The values of current increases abruptly for certain values of θ. The angles θ_1, θ_2 & θ_3 are obtained from the graph. The ratio $\sin\theta_1 : \sin\theta_2 : \sin\theta_3 = 1:2:3$, which is in consistent with Bragg's law.



Crystal Structure of NaCl (Using Bragg's Spectrometer)

In case of NaCl, maximum intensity is obtained for the angles 5.9°, 8.4° & 5.2° respectively using three planes 100, 110 & 111.

$$d = \frac{n\lambda}{2\sin\theta} \quad [n = 1]$$

$$d_{100} : d_{110} : d_{111}$$

$$= \frac{1}{\sin 5.9} : \frac{1}{\sin 8.4} : \frac{1}{\sin 5.2}$$

$$= \frac{1}{0.103} : \frac{1}{0.146} : \frac{1}{0.09}$$

$$= 1 : 0.705 : 1.14$$

$$= 1 : \frac{1}{\sqrt{2}} : \frac{2}{\sqrt{3}}$$

$\Rightarrow$ NaCl is fcc lattice.

Numericals

Q. Calculate the longest wavelength that can be analysed by a rock-salt crystal of spacing 2.82 Å in second order?

Sol^n:

$n = 2$
$d = 2.82$ Å

$2d \sin\theta = n\lambda$

$\Rightarrow \lambda_{max} = \frac{2d}{n} = 2.82$ Å [For maximum $\sin\theta = 1$]

Q. In Bragg's reflection of X-ray, a reflection was found at 30° with lattice planes of spacing 187 Å in second order. Calculate the wavelength of X-ray?

Sol^n:

$\theta = 30°$, $d = 1.87$ Å & $n = 2$

$\therefore \lambda = \frac{2d\sin\theta}{n} = 0.935$ Å.

9) The first order reflection from the plane of NaCl is obtained at $2\theta = 20°$. If 'd' is $2.82\ \mathring{A}$, calculate the wavelength of X-ray used?

Sol^n:

$\theta = 10°$, $d = 2.82\ \mathring{A}$ & $n = 1$

$\therefore \lambda = \frac{2d\sin\theta}{n} = 0.9793\ \mathring{A}$.